

Charlie Marx

CURRICULUM VITAE - SEPTEMBER 2019

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EDUCATION

Haverford College

B.S. Candidate in Computer Science

B.S. Candidate in Mathematics

Minor Candidate in Statistics

2016 – PRESENT

GPA: 3.94/4.0

Relevant Coursework

Machine Learning (TBD*) · Linear Algebra (A) · Probability (A) · Multivariable Calculus (A)

Analysis of Algorithms (TBD) · Programming Languages (A–) · Theory of Computation (A)

Real Analysis (TBD) · Abstract Algebra (A) · Advanced Topics in Probability (A)

*Graduate course at the University of Pennsylvania

AWARDS & HONORS

Beckman Scholars Award

2018

National undergraduate scholarship for excellence in research. Award includes \$26,000 of funding for research and travel.

Brian Kovaric Award Finalist

2017

Faculty-nominated scholarship for one outstanding first-year student.

Farmers Insurance Standout Student of the Year Award

2016

Regional award for academic excellence and community engagement awarded to one student each year.

PUBLICATIONS

MACHINE LEARNING METHODS

[Disentangling Influence: Using Disentangled Representations to Audit Model Predictions](#)

Charles T. Marx, Richard L. Phillips, Sorelle A. Friedler, Carlos Scheidegger, Suresh Venkatasubramanian

Conference on Neural Information Processing Systems (NeurIPS), 2019 - Accepted

[Predictive Multiplicity in Classification](#)

Charles T. Marx, Flavio du Pin Calmon, Berk Ustun

ACM Conference on Fairness, Accountability, and Transparency (FAT)*, 2020 - In Submission

MACHINE LEARNING APPLICATIONS

[Examining Inputs of Biogenic and Oil-Derived Hydrocarbons in Surface Waters Following the Deepwater Horizon Oil Spill](#)

Helen K. White, **Charles T. Marx**, David L. Valentine, Charles Sharpless, Christoph Aepli, Kelsey M. Gosselin, Veronika Kivenson, Rachel M. Liu, Robert K. Nelson, Sean P. Sylva, Christopher M. Reddy

ACS Earth and Space Chemistry, 2019 - Accepted

Rapid Identification of Deepwater Horizon Oil Residues Using X-Ray Fluorescence

Anna P. M. Michel, Alexandra E. Morrison, **Charles T. Marx**, Helen K. White
ACS Environmental Science & Technology Letters, 2018 - Accepted

RESEARCH EXPERIENCE

Research Assistant in Machine Learning

2017 – PRESENT

Haverford College

Developed new methods for detecting indirect feature influence in black box models. Published work involving adversarial learning and disentangled representations.

Advisor: Sorelle Friedler

Research Assistant in Computer Science

2019

Harvard University

Identified new measures for multiplicity in machine learning models. Developed novel measurement tools, and managed experiments on a research computing cluster to show our tools are efficient and accurate on real-world datasets.

Advisors: Berk Ustun and Flavio Calmon

Research Assistant in Chemistry

2017 – 2018

Haverford College

Designed interpretable machine learning tools for scientists to identify oil in the field. Collected oil samples in Alaska and the Gulf of Mexico, and worked with domain experts to optimize model performance and interpretability.

Advisor: Helen White

Summer Research Intern

2018

Prince William Sound Science Center

Worked with oceanographers in Alaska to automate salmon and herring tracking using sonar sensors and machine learning.

Advisor: Scott Pegau

Research Assistant in Digital Scholarship

2017

Haverford College

Analyzed 100,000 handwritten Spanish texts using custom computer vision models.

Advisor: Andrew Janco

ORAL PRESENTATIONS

UC Berkeley Simons Institute for the Theory of Computing

2019

Berkeley, CA

Disentangling Influence (45 minute talk)

Fairness, Accountability and Transparency in Machine Learning

2018

New York, NY

Auditing Black Box Models (60 minute tutorial)

with Carlos Scheidegger and Suresh Venkatasubramanian

Gulf of Mexico Oil Spill & Ecosystem Science Conference

2018

New Orleans, LA

Examining Inputs of Biogenic and Oil-Derived Hydrocarbons (15 minute talk)

POSTER PRESENTATIONS

Beckman National Symposium <i>Irvine, CA</i> Indirect Influence in Machine Learning Models	2019
Beckman Regional Symposium <i>New York, NY</i> Mitigating Social Discrimination in Deep Learning	2018
KINSC Undergraduate Research Symposium <i>Haverford, PA</i> Examining Biogenic Hydrocarbons After the Deepwater Horizon Oil Spill	2017

CONFERENCE & WORKSHOP PARTICIPATION

CONFERENCES

Fairness, Accountability and Transparency in ML (FAT*)	2018, 2019, 2020
Conference on Neural Information Processing Systems (NeurIPS)	2019
Institute for Operations Research and the Management Sciences (INFORMS)	2019
International Conference on Knowledge Discovery & Data Mining (KDD)	2018
Gulf of Mexico Oil Spill & Ecosystem Science Conference (GOMOSES)	2017

WORKSHOPS AND MEETINGS

Summer Cluster on Fairness at UC Berkeley	2019
Functional Ecological Genomics Workshop	2017

FIELDWORK

Alaska Prince William Sound Research Charter	2018
Gulf of Mexico Shoreline Research Tour	2017

ACADEMIC ACTIVITIES

Teaching Assistant in Machine Learning Held office hours and graded assignments for an upper-level undergraduate machine learning course.	2019 – PRESENT
Peer Tutor Tutored students in statistics, linear algebra, and machine learning.	2018 – 2019
Computer Science Faculty Search Committee Member of search committee tasked with hiring a tenure-track faculty member for the computer science department.	2018 – 2019

EXTRACURRICULAR ACTIVITIES

Haverford College Men's Club Soccer Captain.	2018 – 2019
Haverford College Outdoors Club Leader Designed and led week-long backpacking trips for students at Haverford College.	2017 – 2018
Furniture Forward Leader of organization to redistribute furniture to families transitioning out of homelessness.	2015 – 2017